

will cause the option's value to rise by $0.5 \times 0.10 = 0.05$. Consequently, the option will only decline by 0.15, a decline of 0.20 due to a change in the underlying price combined with an increase of 0.05 due to the increase in implied volatility. The option has a *skewed* or *adjusted delta* of -15 .

The inclusion of a volatility skew in a pricing model will affect the calculation of all option risk measures and can greatly complicate a trader's ability to manage risk. For many traders, it may be best to keep things simple, perhaps using a skew model to generate theoretical values while using a traditional model to calculate the delta, gamma, theta, and vega. For an active trader who carries large option positions, calculating accurate skewed sensitivities becomes much more important because the total value of the position can change very quickly as market conditions change. Financial engineers at professional option trading firms are often responsible for developing methods to accurately calculate theoretical values and risk sensitivities using a volatility skew model. But even the most sophisticated model is unlikely to generate values that exactly model option prices under all market conditions. A model can help, but it will always have limitations.

By combining the volatility term structure across expiration dates with the volatility skew across exercise prices, we can form a *volatility surface*. While sometimes difficult to visualize, a volatility surface may enable a trader to more easily see the basic volatility characteristics of an option market. The more exercise prices and expiration dates that are available, the more accurate will be the volatility surface. Sample volatility surfaces for the FTSE 100 options and wheat options are shown in Figures 24-12 and 24-13. At the time, the FTSE 100 Index was trading at 5,966, and the front-month wheat futures contract was trading at 647.

Figure 24-12 FTSE 100 volatility surface, March 16, 2012.

